DLS Buddy — User Manual

Applies to v1.17.0 and later

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1. Introduction

DLS Buddy is a general-purpose Python program for analyzing **dynamic (DLS)** and **static (SLS)** light scattering data. It is not tied to any specific instrument, solvent, or polymer system. The physical parameters — viscosity, refractive index, temperature, $\mathrm{d}n/\mathrm{d}c$, wavelength, and scattering angle — are yours to set: you enter them, or accept and commit the values the solvent library proposes for common solvents (refractive index and viscosity only; $\mathrm{d}n/\mathrm{d}c$ stays hand-entered, and may be left blank — see §6.1).

The graphical interface is organized as a sidebar workspace (your loaded data, auto-grouped into samples) plus tabbed modules:

- **Data** — confirm and edit the physical parameters of each measurement.
- **DLS** — correlation-function fitting (cumulants, exponentials, KWW), CONTIN/NNLS distributions, and multi-angle / concentration-series analysis.
- **SLS** — Zimm, Berry, and Debye analysis for M_w, R_g, and A_2, with a calibration panel, plus **Derived Intensity** and **Mode-Resolved** sub-tabs for a static analysis built from DLS runs (§10.2.1–10.2.2).
- **Cross-Sample** — quantities that link samples or modules: the shape parameter $\rho = R_\mathrm{g}/R_\mathrm{h}$, and log–log scaling plots.
- **Utilities** — data-quality diagnostics (intensity traces, I·sin θ), a synthetic correlogram generator, and the Solvent Explorer (§7.2).
- **Settings** — global defaults and appearance.

A core design goal is to support mixed DLS+SLS sessions: load an R_h measurement (DLS) and an R_g measurement (SLS) of the same sample, and the program pairs them automatically to compute $\rho = R_\mathrm{g}/R_\mathrm{h}$ (§10.3) — no extra step needed.

Launching the program

Double-click the launcher for your operating system:

- **Windows** — `Launch DLS Buddy (Windows).bat`
- **macOS** — `Launch DLS Buddy (MacOS).command` (on first run macOS Gatekeeper may refuse an unsigned script — right-click it → **Open** → confirm once, and it opens normally thereafter)

Or run it from the repository root with:

```
python -m gui.main
```

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2. The Workflow at a Glance

Most analyses follow the same short path. Here is the whole flow in one place; the rest of the guide fills in the detail.

Launch → Load → Confirm parameters + Update → open the DLS or SLS tab → pick a method and Run → read the results → Export.

1. **Launch** the program (`python -m gui.main`, or the double-click launcher for your OS).

2. **Load** your data — **Load DLS correlogram...**, **Load SLS intensities...**, or **Load count-rate trace...** — the program recognizes the common instrument formats and groups the files into **samples** in the left sidebar (§3).
3. **Confirm the parameters** on the **Data** tab — temperature, viscosity, refractive index, wavelength, scattering angle, and (for SLS) concentration and $\mathrm{d}n/\mathrm{d}c$ (blank is allowed — §6.1). Identity and optics are entered **once per sample**; press **Enter** to jump to the next field. Click **Update** to commit — analysis runs on the committed values (§6).
4. **Open the analysis tab** for what you need:
   - **DLS** (size, R_h): tick the measurement(s) in the sub-tab's list and choose a method — **Correlogram** (cumulant / exponential) or **Distribution** (CONTIN / NNLS / Lognormal) — then **Run**. The multi-angle Γ **vs** $\mathbf{q}^2$ and concentration-series **D vs c** tabs tick the points to include in the fit (§10.1).
   - **SLS** (M_w, R_g, A_2): select the sample, set the **calibration**, choose **Zimm / Berry / Debye**, then **Run** (§10.2).
5. **Read the results** — each method fills a results **table** plus a plot; flags mark *apparent* vs *thermodynamic* and *uncalibrated* results so they aren't over-read.
6. **Export** with **Export CSV...** (Origin-compatible) (§11).
7. *(Optional)* Combine a sample's DLS R_h and SLS R_g on the **Cross-Sample** tab for $\rho = R_\mathrm{g}/R_\mathrm{h}$ and the scaling plots (§10.3).

---

3. Loading Data

The program reads data from several instruments plus a generic plain-text fallback (§3.1) into one common format, so measurements from different sources load into the same workspace and analyze the same way.

Every file load follows the same workflow:

1. Click **Load DLS correlogram...** (or **Load SLS intensities...**, or **Load count-rate trace...**) and pick the file(s) — you can select **several at once**, and each is auto-detected independently (any that can't be read are reported at the end, the rest still load). The program **auto-detects** the format, trying each supported instrument parser until one recognizes the file.
2. The parser reads the file and pre-fills any parameters it can find.
3. The **Data** tab shows all fields. Change any pre-filled value, and fill in whatever your analysis needs — which fields are required depends on the analysis (e.g. concentration is required for SLS but optional for basic DLS sizing, §6).
4. **Update** commits the parameters; analysis then runs on the committed values.

   Note. Pre-filled values from a file are suggestions, not locked inputs. You can always override what the parser found — for example to correct an instrument software value or to apply a calibration you determined independently.

   Tip. To set a value on several measurements at once, Ctrl- or Shift-click them in the sidebar, edit the focused one, and press **Update** — your edits apply to every highlighted measurement. (Identity and optics already propagate across a whole sample automatically.)

3.1 Supported File Formats

The load buttons carry no vendor name: they first try to auto-detect among the instrument formats below (each sniffs its own format). If none match, the program falls back to a **plain two-column** file — for a generic DLS file it asks for the delay-time unit and data form (g_2-1 / g_2 / g_1); a generic SLS file (angle, intensity) needs no prompt. Only if that also fails does it report an unrecognized format.

Format	Produces
Brookhaven Particle Explorer DLS `.CSV`	One DLS measurement per file
Brookhaven raw ASCII DLS `.dat`	One DLS measurement per file, with angle, wavelength, temperature, viscosity and refractive index pre-filled from the file (see §5.2)
Brookhaven Particle Explorer count rate `.CSV`	One intensity trace per file
Brookhaven Particle Explorer SLS `.CSV`, concentration series	One measurement per concentration (solvent reference included)
Brookhaven Particle Explorer SLS `.CSV`, single angular scan	The sample — plus the solvent blank when the run recorded one beside it (see §5.4)
ALV correlator `.ASC` (e.g. ALV-7012 CGS-12F)	**All angles** in the file as separate DLS measurements (one sample) + a count-rate trace per angle
Malvern Zetasizer clipboard correlation text	**All records** in the file as separate DLS measurements (no metadata — all parameters entered at confirmation); a record column with a header but no data is skipped, and the skipped records are listed at load
Malvern Zetasizer structured *export* `.txt`/`.csv`	**One DLS measurement per row**; pre-fills refractive index, temperature, viscosity, and the sample/material/dispersant names (see §5.5)
NanoTemper Prometheus Panta `.xlsx` autocorrelation export	One DLS measurement per file (no metadata — all parameters entered at confirmation); the repeated correlator channels at each multi-tau block boundary are discarded at load (see §5.6)
Plain-text two-column DLS `(delay, g₂−1/g₂/g₁)`	One DLS measurement per file (fallback; prompts for delay unit + data form)
Plain-text two-column SLS `(angle, intensity)`	One concentration per file (fallback; loads directly)
Plain-text two-column trace `(time, count rate)`	One intensity trace per file (fallback; prompts for time + count-rate units)

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4. Generic (Plain-Text) File Formats

The generic parsers accept data from any source — home-built correlators, simulation output, collaborator data, spreadsheet exports — as long as it is a clean two-column numerical table. Because these files carry no metadata, every physical parameter is supplied in the Data tab.

4.1 Generic DLS Format

File contract (the parser rejects non-compliant files):

- Exactly two columns of numbers; no header, comment, or blank lines.
- Comma- or tab-separated (auto-detected).
- Column 1: delay times (unit specified at confirmation).
- Column 2: correlation values (interpretation specified at confirmation).
- At least two data rows.

Delay-time units: `s`, `ms`, `us` (µs), `ns` — converted to seconds on load.

Column-2 interpretation (data form). The program converts every input to $g_2(\tau)-1$ (baseline-subtracted intensity autocorrelation) before storage:

Data form	Column 2 contains	Extra parameter
`g2m1`	$g_2(\tau)-1$, already baseline-subtracted	none
`g2`	$g_2(\tau)$, baseline not yet subtracted	baseline $B > 0$
`g1`	field autocorrelation $g_1(\tau)$	coherence factor $\beta \in (0, 1]$

For the `g2` form, the baseline is removed; for the `g1` form, the Siegert relation converts the field autocorrelation to the measured quantity (β is the instrument coherence factor, an optical property, typically 0.01–1.0 for standard goniometers, determined independently).

> **Tip.** Unsure which form you have? Check the range of column 2. $g_2(\tau)-1$ starts near the intercept β (often 0.3–1.0) and decays to 0; unsubtracted $g_2(\tau)$ starts near $1+\beta$ and decays to 1; $g_1(\tau)$ starts near 1 and decays to 0.

Because g_1 is *normalized*, the program checks that a column declared `g1` peaks between 0.01 and 1.05 and refuses it otherwise, naming the peak it measured. A column of raw counts, a column in per cent, or an unsubtracted g_2 from an ordinary instrument all land outside that range. The one case the check cannot settle for you is an unsubtracted g_2 from a very low-coherence instrument (β below about 0.05), which peaks close enough to 1 to look like g_1 — so if the baseline has not been subtracted, choose `g2` and supply B.

4.2 Generic SLS Format

Each file is one concentration at multiple angles (one file per concentration).

- Exactly two columns; no header/comment/blank lines; comma- or tab-separated.
- Column 1: scattering angles in degrees, each strictly in (0, 180).
- Column 2: raw intensity values.
- At least two data rows.

> **Column order.** Angle must be column 1, intensity column 2. If reversed, the parser detects column-1 values outside (0, 180)° and raises an error suggesting the columns may be swapped.

For a Zimm/Berry/Debye dataset, load one file per concentration (including the solvent reference at $c = 0$) and set each concentration in the Data tab. The SLS module assembles the dataset from the loaded measurements.

5. Instrument File Formats

The vendor parsers read each instrument's native export. The formats are described for reference; the parsers handle all reading automatically.

5.1 Brookhaven DLS Correlation Function (`.CSV`)

- Row 1: sample label string (e.g. `PnVP (40k) in Water - 6/1/2026 4:46:59 PM`).
- Row 2: column headers (`t(µs)`, `C(t)`, `Fitted (CONTIN)`).
- Rows 3+: delay time (µs), $g_2(\tau)-1$, and the instrument CONTIN fit.
- Trailing padding rows with $t = 0$ and $C(t) = -1$ fill unused correlator channels; they are stripped automatically.

  Important. The DLS file contains no instrument metadata. Angle, wavelength, refractive index, temperature, viscosity, and sample name are all supplied manually in the Data tab. **Concentration is optional for DLS** — cumulant, exponential, KWW, and CONTIN/NNLS size analysis (and Γ vs q^2) do not use it; it is needed only for the multi-concentration diffusion extrapolation.

  Note. The program ignores the Brookhaven CONTIN fit column. All regularization is performed independently by the program's own pipeline.

The correlogram and count-rate history are separate files and are associated manually. (Sample labels in the two files sometimes differ slightly in spacing — a known Brookhaven quirk — which is why automatic pairing is not attempted.)

5.2 Brookhaven Raw ASCII DLS Export (`.dat`)

The same software also exports a raw ASCII file, described in Appendix A ("Raw Data File Formats") of the Brookhaven TurboCorr correlator manual. It is a different format from the `.CSV` above, and a more complete one.

- Lines 1–37: one number per line, a fixed list of correlator and sample parameters. Among them are the angle, wavelength, temperature, viscosity and refractive index — the values that must be typed in by hand for the `.CSV`.
- Then one row per correlator channel: delay time (µs) and the **raw channel counts**, not $g_2(\tau)-1$.
- Then four closing lines: sample ID, operator, date, and time.

Because the channels hold raw counts, a baseline is needed before there is a correlogram at all: $g_2(\tau)-1$ = counts / B − 1. The file carries two candidates. The **calculated** baseline comes from the run totals (counts A × counts B, over the number of samples), so it reflects the whole measurement. The **measured** baseline is the average of the baseline channels the acquiring software set aside. The program uses the calculated baseline unless the file names one itself; Settings ▶ File Loading changes that default. The two normally differ by a few tenths of a percent, and for most work that difference does not reach the answer: the default cumulant fit and all the distribution fits float or estimate a baseline of their own, which absorbs it, so the recovered size is the same either way. The exception is the **linear** cumulant method, which assumes the baseline is exactly

zero; there the choice shifts the size by a few percent. A large difference between the two baselines is worth noticing whichever you pick — it is a sign of dust or a slow drifting mode.

Sample identity — polymer, solvent, concentration — is not in the file and is entered in the Data tab, as for every other format.

5.3 Brookhaven Count Rate History (`.CSV`)

- Row 1: sample label string.
- Row 2: column headers (`Time (seconds)`, `Count Rate (kcps)`).
- Rows 3+: elapsed timestamp (`HH:MM:SS.sssssss`) and count rate in kcps.

Timestamps are elapsed time from the start of the run, not wall-clock time. Count rates are converted to counts per second (cps) on load.

5.4 Brookhaven SLS Intensity Export (`.CSV`)

The instrument writes SLS intensities in two layouts. Both are flat lists of key–value pairs sharing a metadata header, and the program tells them apart on its own — the same *Load SLS intensities…* button reads either.

Concentration series (the Zimm or Debye plot export). One file holds every concentration and angle: a metadata header, an angle table, a concentration table, and an intensity grid (`Intensity - Concentration N - Angle M, value`). The parser extracts all available metadata. Two fields are absent and are entered manually:

- Sample name.
- Temperature (the SLS export has no temperature field).

Single angular scan (the export for a one-concentration experiment, such as a Guinier measurement). One file holds one acquisition as `Sample Intensity - Angle N` rows, with no concentration table. The instrument's plain export — a sample scan, a solvent scan, the toluene calibrant — loads as **one** measurement. When a blank was measured in the same run the file also carries `Liquid Intensity - Angle N` rows, and it then loads as **two**: the sample, and the blank at c = 0, which the program marks **solvent ref** in the sidebar. Load several such scans into one sample and each series is referenced against the blank from **its own file** — a blank drifts with cell cleanliness, alignment and vat temperature between acquisitions, which is why the instrument records one per run. Every blank beyond the reference is kept; the sidebar marks it **extra blank** because it is not the sample reference, not because no series uses it; a series whose file carried no blank (the plain export, a separate-file blank, or a generic loader) uses the sample's reference blank (the first one loaded) unless you pick one in the Data tab's **Solvent blank** row. Four fields are entered manually — the two above, plus:

- Concentration (this layout records none) — enter **0** for a solvent or calibrant scan, which files it as the solvent reference of its sample.
- $\mathrm{d}n/\mathrm{d}c$ (this layout carries no $\mathrm{d}n/\mathrm{d}c$ row) — or leave it blank, §6.1.

This layout has no calibration block either, so Mw and absolute A_2 need a calibrant point entered in the SLS tab, exactly as for an uncalibrated concentration series. Rg is unaffected.

> **Rayleigh ratio.** The Brookhaven software embeds its own toluene Rayleigh ratio. It is read and recorded for traceability but not used in analysis — the program computes the

temperature- and geometry-corrected value from the toluene Rayleigh literature. See the **Theory-and-Equations-Guide** §1.5.

5.5 Malvern Zetasizer export template

The Zetasizer software's **export** is configurable, so the program reads its columns by **name** (not by position) and accepts any column order. To produce a file the export parser recognizes, configure the exporter as follows:

- **Settings:** include the header row; use commas as the separator.
- **Parameters (any order):** Sample Name, Material Name, Dispersant Name, Dispersant RI, Temperature, Viscosity, Correlation Delay Times, Correlation Data.

The two correlation blocks (Delay Times + Data) are required; each named parameter is optional and pre-filled when present. The dispersant name is taken **exactly as the instrument recorded it** (§7) — it is free text typed by whoever ran the sample, so the program does not reinterpret it. The program maps the columns as:

Export column	Pre-fills
Sample Name	sample label
Material Name	polymer/material name
Dispersant Name	solvent name
Dispersant RI	solvent refractive index
Temperature	temperature
Viscosity	viscosity
Correlation Delay Times	lag-time axis
Correlation Data	correlogram (g_2-1)

Scattering **angle**, **wavelength**, and **concentration** are not in the export and are entered at the confirmation step, as with any DLS file.

5.6 NanoTemper Prometheus Panta autocorrelation export (`.xlsx`)

The Panta writes its autocorrelation function to a spreadsheet with two worksheets: a `Chart` sheet, which the program ignores, and an `ACF` sheet holding the data under the headers `Tau [µs]` and `Autocorrelation`. The delay unit is read from that first header rather than assumed, so an export written in different units still loads correctly.

Two things about this format are worth knowing.

The file carries no measurement parameters. There is no scattering angle, wavelength, refractive index, temperature, viscosity, concentration or sample name anywhere in it, so every parameter is entered at the confirmation step and nothing is pre-filled. The scattering angle deserves particular care: it enters the recovered size through q^2, so an angle entered wrongly rescales every result from the file without anything looking out of place. Take it from your instrument's documentation rather than from memory.

Repeated delay times are expected, and are removed on load. The Panta uses a *multi-tau* correlator, which covers a very wide range of delay times by doubling its sampling interval every block of channels — the only practical way to span nanoseconds to seconds in a few hundred

channels. Each block boundary is reported twice, once by the finer block and once by the coarser one, so the same delay time appears on two rows with slightly different correlation values.

The program keeps the finer block's value and discards the other, which is what the multi-tau scheme itself prescribes (the source is listed in the Citation-Index). A note at load says how many channels were removed. Keeping the other value instead was measured on a real Panta file to shift the z-average size by 0.02% and to leave the recovered size distribution's peak unchanged — against 0.5–17% for the ordinary analysis choices you make yourself on the same file, so which of the two is kept has no bearing on any result you would report.

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6. Physical Parameters and Units

You enter and read each quantity in whatever unit is convenient. In the **Data** tab each parameter row has a **Unit** column: a dropdown for quantities with a choice of units (the value re-displays when you switch units — the underlying value is unchanged), or a fixed label for quantities with a single natural unit.

The identity and label fields carry no units; the physical quantities do. The **Purpose** column notes what each field feeds:

Parameter	Unit or control	Input units (default first)	Purpose
Label	— (text)	—	Free-text tag for your own reference; does not affect grouping or analysis
Solute name	— (text, case-insensitive)	—	Names the scatterer (need not be a polymer); part of sample identity
Solvent name	— (dropdown + **Other...**, case-insensitive)	—	Part of sample identity; picks library n/η for common solvents (§7.1)
M_w fraction	— (text, case-sensitive)	—	Optional label separating molecular-weight fractions of one polymer (§8.1)
Polarization (DDLS)	— (VV/VH/VU)	—	Tags a correlogram's polarization for depolarized DLS (DDLS) analysis (§10.1); blank = ordinary DLS
Chain type	— (Linear/Branched)	—	Optional architecture tag for the Cross-Sample branching factors g and h (§10.3): **Linear** = a reference defining the linear R_g–M_w / R_h–M_w laws; **Branched** = a target that gets g / h. Shared across the sample; blank = untagged

Parameter	Unit or control	Input units (default first)	Purpose
Solvent blank	auto (or one of the sample's blanks)	—	SLS only, on a $c > 0$ series: the $c = 0$ measurement subtracted from this series. **auto** = the blank recorded in this series' own file, else the sample's reference blank; pick one when the file carried none or the automatic pairing is wrong. The dropdown tooltip says which blank is in use and why (§5.4)
Concentration	g/mL	**mg/mL**, g/L, µg/mL, g/mL	Required for SLS and the DLS concentration series
Temperature	kelvin (K)	**°C**, K, °F	Sets Stokes–Einstein R_h and library properties
Viscosity	Pa·s	**mPa·s**, cP, Pa·s	Solvent viscosity η for Stokes–Einstein
Solvent refractive index	— (dimensionless)	—	Sets the scattering vector q; library-proposed for common solvents (§7.1)
Wavelength	nanometers (nm)	nm	Laser wavelength; sets q
Scattering angle	degrees	deg	Detector angle; sets q
$\mathrm{d}n/\mathrm{d}c$	mL/g	mL/g	Refractive-index increment (SLS); the most sensitive SLS input. May be left **blank** (§6.1); zero is refused
Mean count rate (DLS)	counts/second (cps)	**cps**, kcps, Mcps	Average count rate for the run; proposed from a linked trace or the instrument file, never overwritten once you type one (§9.1)
Delay-time window (DLS tab)	seconds (s)	**µs**, ns, ms, s	Restricts which lags are fitted
Count rate (trace)	counts/second (cps)	**kcps**, cps, Mcps (chosen on load)	Intensity-trace units

SLS intensities. The calibrant intensity and dark count rate are entered in **the same units as your SLS intensity file** (whatever the instrument exported). The program does not convert them, because only the *ratio* of calibrant to sample intensities enters the Rayleigh ratio — a fixed-unit conversion there would introduce a scale error. The fields are labeled "(file units)" accordingly.

6.1 When you have no dn/dc

The refractive-index increment is the one SLS parameter that is sometimes genuinely unavailable: no value has been published for your polymer in your solvent, or you are measuring a particle dispersion, for which the quantity is not defined at all.

Leave the cell blank. Every analysis still runs. What you get is:

- **Unaffected** — every R_g, the form-factor sizes and axial ratio, and the calibration-free $2 \cdot A_2 \cdot M_w$. These come from *ratios*, and the unknown optical factor cancels out of a ratio exactly.
- **On an arbitrary scale, and flagged as such** — the absolute M_w and A_2. They are dashed out at the value, with a flag naming dn/dc as the reason.

This is the same treatment an **uncalibrated** run gets, and the two are reported separately: if a value is dashed, the label tells you whether to calibrate, to enter a dn/dc, or both. Calibrating will not repair a missing increment, so the program never implies that it would.

> **Do not type 0 to get past the field.** It is refused, and for a reason: zero is not "unknown", it is the claim that the sample does not scatter differently from its solvent, which makes every fitted quantity meaningless. Blank is the honest entry. A **negative** dn/dc is fine — the optical constant depends on the square, so the sign cancels, and a solute less refractive than its solvent really has one.

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7. Solvent Names

Solvent identity is part of sample identity: the same polymer at the same concentration and temperature in two solvents is two distinct datasets (enabling the comparison of the same polymer in different solvents).

A name is stored **exactly as you enter it** and matched **ignoring capitalization and surrounding spaces — nothing else is normalized**. `Water`, `water` and `WATER` are one solvent; `H2O` is a different one. The program does not decide that an abbreviation stands for a particular solvent, because it has no way to be right about that — a name you type is a name you own, and `My Phosphate Buffer` is as valid an entry as `toluene`.

> **Note.** The rule is the same for the solute name and the solvent name: both ignore capitalization and surrounding spaces, and neither is interpreted any further. `PVP` and `pvp` are one sample.

7.1 Automatic refractive index and viscosity

For common solvents the program can **propose** the two optical/hydrodynamic properties every scattering calculation needs — the **refractive index** n (which sets the scattering vector) and, for DLS, the **viscosity** η (which sets the Stokes–Einstein radius). In the **Data** tab the *Solvent name* row is a dropdown listing every library solvent in two sections — **Measured properties** and **Estimated properties** — followed by **Other...**.

Pick a solvent from **Measured properties** and n (and η) fill in automatically from the sample's temperature and wavelength, each marked with a small **teal dot**.

- The values **re-derive** whenever you change the temperature or wavelength.
- Each proposed value is **rounded at the last digit the library can stand behind**. The uncertainty itself only chooses that place; it is never used in any analysis.
- A value you **type by hand is never overwritten** by the library, and shows no dot (to return to the library value, clear your entry and re-pick the solvent).
- Outside a solvent's validity range the program **does not guess**: it clears the library value and shows a note asking you to enter it manually.

Solvents under **Estimated properties** are listed but **not** auto-filled — their underlying data is too weak to put a number into an analysis unasked. Picking one sets the name, leaves n and η blank, and shows a note saying so; you enter both yourself. Use the Solvent Explorer (§7.2) to see what the library's estimate would be.

Other... is for a solvent the library does not carry. It opens a text box beside the dropdown; the name you type is stored exactly as typed and used as-is to match this measurement to its sample, and n and η are yours to enter. A solvent named this way is its own sample — the program will not merge it with a library solvent it happens to resemble.

The measured-properties solvents and their validity ranges are tabled in §7.3; the full list, with the uncertainties and sources, is in the **Theory-and-Equations-Guide** (§5).

7.2 The Solvent Explorer (Utilities ▶ Solvent Explorer)

The **Solvent Explorer** — a sub-tab of the **Utilities** tab — is a standalone look-up and visual calculator over the whole library. It is global — it needs no loaded sample, ignores the sidebar selection, and never writes into a measurement. Choose a solvent, temperature, and wavelength to read n and η with their uncertainty and a **confidence-tier badge** reading **primary** or **estimate** — the same two tiers as the Data tab's **Measured properties** and **Estimated properties** sections (all library solvents appear here, grouped the same way). Two plots, side by side on one shared figure, show at a glance the range of conditions the library can serve:

- **η and n vs temperature** (twin axes — viscosity on the left, on its own log-capable scale; refractive index on the right), and
- **n vs wavelength**,

each drawn **only across the range the library covers**, with a shaded **confidence band** and a marker at your chosen condition. The band is a conservative bound **at each condition**, including an allowance for how much independent laboratories' published measurements typically disagree — narrowest near the conditions the sources measure best, growing toward the edges of the validity range, and honestly flat where a source states no temperature shape (see the **Theory-and-Equations-Guide** §5). Like every library uncertainty it is **displayed, never folded into a reported ±**. Set which solvent the Explorer opens on under **Settings ▶ Solvent Library ▶ Default solvent**.

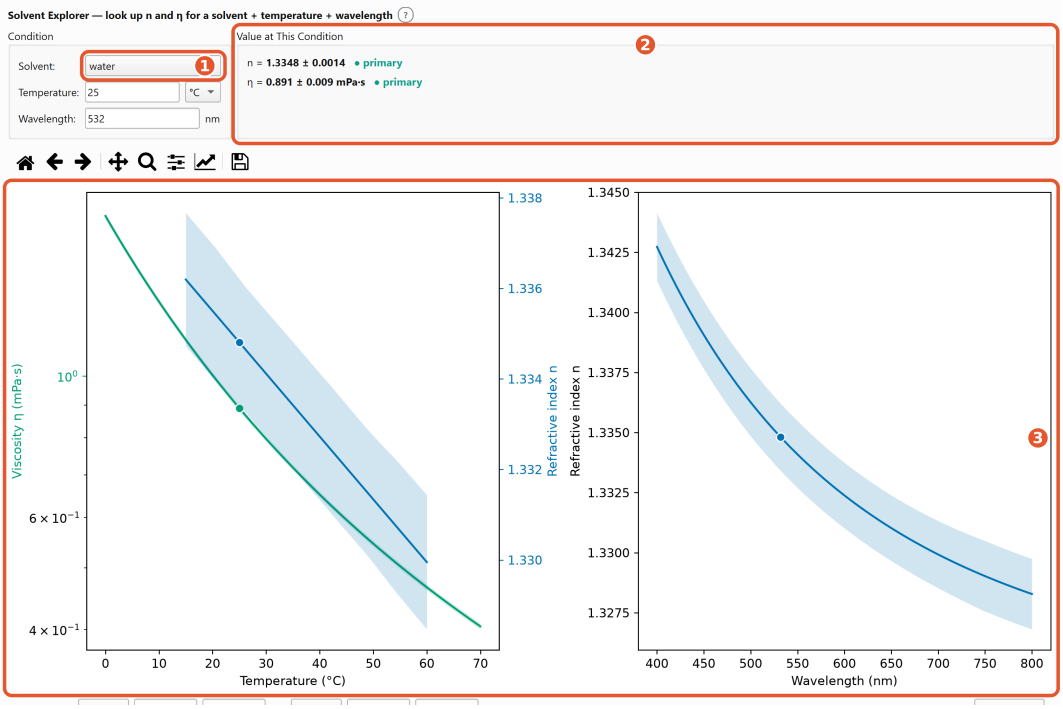


Figure 1: The Solvent Explorer (Utilities ▶ Solvent Explorer): pick a solvent (1); read n and η at the chosen temperature and wavelength with their uncertainty and a confidence-tier badge (2). The plots (3) draw η and n vs temperature and n vs wavelength, each with a shaded confidence band, only across the range the library covers.

7.3 The Solvent Library at a Glance

The table below lists the **primary**-tier solvents — the higher-confidence set that auto-fills n and η in the Data tab (§7.1) — with the range of conditions each one covers. Ranges are **per-solvent**, not uniform: they follow the span of the underlying measurements, so refractive index and viscosity may be valid over different temperature windows for the same solvent. Outside a listed range the program stops proposing a value and asks you to enter it by hand (§7.1). The uncertainties behind these ranges, and the sources they come from, are in the **Theory-and-Equations-Guide** §5.

Solvent	RI λ-range (nm)	RI T-range (°C)	η T-range (°C)
acetone	476–633	15–40	−18–46
benzene	476–633	15–40	15–66
cyclohexane	476–633	15–40	15–76
ethanol	476–633	15–40	−20–70
ethylene glycol	400–800	1–45	0–71
glycerol	400–800	1–45	15–51
hexane	400–640	15–40	−3–66
methanol	450–700	15–50	0–70
toluene	405–830	15–40	−13–96
water	400–800	15–60	0–70

The library also carries 9 **lower-confidence (*estimate*-tier)** solvents, drawn from thinner or less consistent source data: acetonitrile, ccl4, chloroform, cs2, dcm, dmf, dmso, ethyl acetate, and thf. They appear in the Solvent Explorer (labeled *estimate*) but are held back from the Data-tab auto-fill, so a value only ever reaches an analysis when you pick it there yourself.

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8. How the Program Organizes Data

Every measurement carries a sample key — (polymer, solvent, concentration, temperature) — used to associate related datasets, for example to pair a DLS and an SLS measurement of the same sample when computing $\rho = R_g/R_h$. For grouping into a **sample**, the identity is (polymer, solvent, rounded temperature); concentration and angle are axes *within* a sample.

The **Workspace list** on the left is your navigator and data manager: click a measurement to load it into the active tab, click a sample or DLS/SLS heading to select everything under it, and right-click to remove, move/regroup, or average replicates. It only *navigates* — each analysis tab decides what it actually analyzes (you **tick** measurements, or pick a sample from the tab's own **Sample** dropdown). The list also **mirrors** the active tab's current selection — those measurements read **bold blue** — and flags a **not-yet-confirmed sample in amber** (set its identity in the Data tab and press *Update*).

Stored results **follow the sample, not its name**: correcting a typo in a sample's polymer, solvent, or temperature (or moving all its measurements into another sample) carries the stored M_w, R_g, A_2 and R_h values along to the regrouped sample. When a regroup would **merge** two samples that both hold results, each quantity keeps the higher-provenance value (hand-entered beats picked beats computed, and the surviving sample wins a tie) — and the program asks first whenever a hand-entered or picked value would be dropped; *Cancel* applies nothing.

8.1 Molecular-weight fractions

A scaling study measures several **molecular-weight fractions** of one polymer (same chemistry, solvent, and temperature), each with its own concentration series. Because molecular weight is *not* part of the sample identity, all such fractions group into one sample. To keep them separate, give each measurement an optional free-text M_w fraction label (e.g. `250k`, `1M`) in the Data tab; editing it offers to apply the label to every concentration loaded from the same file. The label is *not* required — leave it blank for single-fraction samples or systems with no meaningful M_w (e.g. nanoparticles).

Tag before you fit. If you load more than one molecular weight of the same polymer and leave them untagged, their concentration series share one fraction and an SLS fit pools them into a single meaningless mixed-M_w result. The SLS tab shows a passive note when it spots the signatures of this (repeated concentrations, or more than one solvent blank) — tag each series with its own M_w fraction to fit them separately (or ignore the note if they are replicates of one molecular weight).

Within a sample, the fraction label **partitions the analyses**: the SLS tab shows a *Fraction* selector and runs an independent Zimm/Berry/Debye fit for the selected fraction, while the c = 0 solvent blank follows each series (its own file's blank, else the sample reference — the Data tab's **Solvent blank** row) regardless of fraction. The DLS tab partitions the same way, by the focused measurement's own fraction: the Γ vs q^2 and D vs c tables gray out rows of any *other* fraction (focusing an untagged run fits only the untagged series), so a single fit never mixes fractions. Each fraction therefore yields its own M_w, R_g, A_2 (and, from DLS, R_h). In the Cross-Sample tab each

fraction contributes its own point to the scaling plots (R_g–M_w, A_2–M_w) and its own row in the ρ table, so a molecular-weight series produces a proper scaling curve rather than a single merged point. (The per-fraction Cross-Sample views build on the per-fraction SLS analysis described above.)

Intensity traces (count-rate histories) are independent objects: they can be loaded and analyzed without a correlogram, or linked to a DLS measurement after the fact — supporting batch diagnostics over a concentration series.

8.2 Sessions

The program's persistence is a single autosave, controlled by **Settings ▶ Session ▶ Reopen last session on startup** (off by default) — there is no separate File ▶ Save or Open.

When it is on, closing the program saves the whole workspace to a file the program manages for you, and the next launch reopens it automatically: every loaded measurement and trace, the sample results, the calibration (including any not-yet-committed edit), the SLS masks, and the DLS tab's Mode Map picks and recorded mode sets (§10.2.2) all come back exactly as you left them.

If that file is missing or damaged, the program starts empty. If it was saved by a different version of the program, it cannot be opened: you are asked at startup whether to keep the unreadable file (nothing is auto-saved over it) or let this version replace it the next time you quit.

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9. Data Quality and Utilities

The Utilities module assesses data quality (especially for intensity traces) and generates synthetic test data. These help identify slow-mode fluctuations, drift, and dust before committing to full correlation-function or Zimm analysis.

9.1 Intensity-Trace Diagnostics

Where traces live. A count-rate trace is loaded from **Load count-rate trace...** in the Workspace panel, beside the DLS and SLS load buttons. An ALV `.ASC` loaded as a correlogram brings its own trace with it, already attached to that measurement — attached traces appear nested under their run in the Workspace list, unattached ones in a **Traces** group at the bottom. **Right-click a trace** to attach it to a DLS measurement, detach it, export it as `(time, count rate)` columns, or remove it. A measurement carries at most one trace, so attaching a second detaches the first (which becomes standalone — it is not deleted). Removing a measurement deletes the trace attached to it, and the confirm says how many before it happens.

The **Utilities ▶ Traces** tab chooses which traces are *plotted*; it does not own membership.

Mean count rate. A trace attached to a DLS run proposes that run's **Mean count rate** on the Data tab, labeled with a teal dot. The instrument file's own value is offered as an alternative where the format records one. As with every proposed value, it is freely editable and a value you type is never overwritten — not by re-loading, not by re-deriving. If your value disagrees with the linked trace by more than 5 %, a note below the table says so; the value is still used exactly as you entered it.

- **Summary statistics & baseline.** Mean, standard deviation, coefficient of variation, min, max. The baseline (true scattering level) is estimated robustly so upward dust spikes do not

bias it — by a lower percentile (default 25th) or a sigma-clipped mean.

- **Outlier flagging.** Samples that stand out from the rest of the trace are flagged. The band is centered on the trace's **median** and scaled by the trace's **own** scatter — 1.4826 × the median absolute deviation — and a sample is flagged when it falls outside $\pm k$ of that (default $k = 3.5$). Both choices resist the dust being looked for: a median center is not dragged by a handful of spikes, and a scatter-based scale adapts to traces that are noisier than photon statistics alone. The flagged fraction is a compact quality metric.

  Interpretation — this asks a narrower question than it looks. The flag compares each sample against *this trace's* scatter, so a trace can be far noisier than photon statistics permit and still flag almost nothing: every sample is unusual, so none stands out. Whether the trace as a whole is too noisy is the **Fano factor**'s (variance/mean of the counts; 1 for pure shot noise) question, reported in the same summary line — read the two together. The band is also global, so it flags departures from the baseline of any duration without telling you how long they lasted; a slow drift trips it much as a single spike does. **Windowed Hampel**, below, is the overlay that separates those two.

  A trace with no measurable scatter cannot be judged, and says so. If more than half the samples are *identical* — a stalled or blocked detector, or a very coarsely quantized reading — there is no scatter to measure against, and the overlay reports **no usable scale** instead of flagging. Nothing is flagged in that state, including an otherwise-obvious spike: with no scale there is no basis to call any sample unusual. Read the trace plot and the Fano factor directly when you see it.

- **Windowed Hampel.** The same test against a **moving** baseline, which is what lets it say *when* something happened. Over a window of w points centered on each sample it takes the median and the same robust scatter, draws that rolling median with its $\pm k$ band, and marks the samples that leave it. The two overlays share one k, so they differ only in what they compare against. The summary line adds an **event** count — consecutive flagged samples grouped into one event, with its duration — so a single dust spike and a five-second excursion are no longer the same number.

  Sizing the window is the whole skill, and the trade-off is real. An event lasting up to about **half the window** can be flagged; anything longer is tracked *as* baseline. So a persistent offset — dust that settles and sits, as it does in a viscous solvent — appears as a **step in the rolling curve** with no flags at all. That is a property of the method, not a fault, and it is why the curve is drawn and not just the flags: what the layer absorbs, you can still see. Widen the window to catch longer events; narrow it to follow a faster-moving baseline. A narrow window also flags more of a perfectly clean trace (measured on synthetic data: about 1.6 % of samples at 11 points against 0.1 % at 101, for $k = 3.5$), so prefer the widest window that still resolves the events you care about. Leaving it on **auto** sizes it from the trace length; the value used is shown in the summary line and written into the export.

  A gap in the band means that stretch was not judged. Two causes, both reported as percentages in the summary line: no full window fits (the first and last half-window of every trace), or more than half the window shares one value, leaving

no scatter to measure — the same condition the global band reports as *no usable scale*, applied per window. Nothing is flagged inside a gap, and no scale is invented to fill it. The other diagnostics still cover those samples.

Use **Export decomposition...** to write the whole thing out as columns — time, count rate, rolling median, rolling scale, the flags, and where the layer abstained — with both overlays' settings in the header. The trace itself is exported unaltered: this diagnostic flags, it never filters.

Which trace is "focused"? With several traces ticked, the summary line, the Diagnostic sub-plot and every **Export ...** button describe one trace: the highlighted one if it is among the ticked traces, otherwise the first ticked one — the summary line starts with its name. Ctrl-clicking a trace off the overlay leaves it highlighted but no longer ticked, so the exports move to the first ticked trace with the summary line, never to the trace you just removed.

- **Running average.** Sliding-window mean and standard deviation (time- or point-based window). The time-based window handles irregular or non-uniform sampling; a wide window (30–60 s) separates slow drift from shot noise.
- **Block-variance (blocking) analysis.** The standard error of the mean vs. block size: flat for an uncorrelated trace; a rise-then-plateau is the signature of positive correlations (slow modes). The full curve is returned with a correlation flag (robust plateau-vs-small-block ratio, default threshold 1.5 — a general heuristic, not a system-specific cutoff).
- **Count-rate histogram.** Fitted with a Gaussian and/or Poisson distribution (Freedman–Diaconis binning, Sturges fallback). The key diagnostic is the **Fano factor** (variance/mean in count space): ≈ 1 is shot-noise limited (ideal Poisson); > 1 signals excess fluctuations (slow modes, dust, drift). Reduced χ^2 of each fit and the coefficient of variation (vs. the shot-noise expectation $1/\sqrt{N}$) are also reported. See the **Theory-and-Equations-Guide** for the equation.
- **Stationarity test.** An augmented Dickey–Fuller test gives a p-value for whether the trace is stationary; non-stationarity indicates drift or slow-mode behavior.

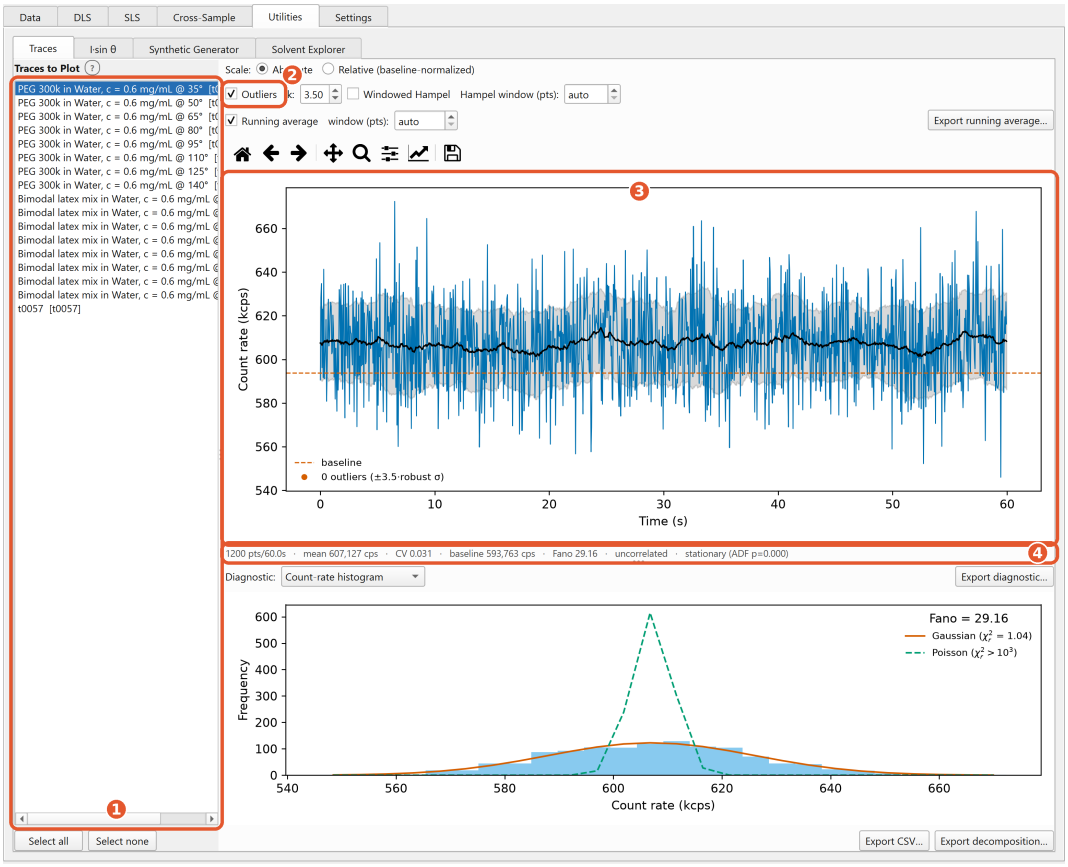


Figure 2: Utilities ▶ Traces. Tick the trace(s) to plot (1) and toggle the overlays (2). The count-rate-vs-time plot (3) shows the robust baseline (dashed), flagged outliers (dots) and the running average (solid) with its ±1 SD band; the summary line (4) reports the point count, mean, coefficient of variation, baseline, and the Fano factor. The **Diagnostic** combo below it picks the second plot — here the count-rate histogram with its Gaussian and Poisson fits. Traces are loaded, attached and removed in the Workspace panel, not here.

9.2 SLS I·sin θ Diagnostic

For an ideal, isotropic, dust-free scattering volume, I·sin θ is flat across angle. Curvature, asymmetry about 90°, or upturns at the extremes reveal alignment problems, stray light, or dust. Curves can be shown **absolute** (comparing concentrations) or **normalized** (each divided by its mean — for comparing solvents, e.g. toluene vs. water, where absolute scale is meaningless). Choose which sample to plot from the **Sample** dropdown at the top of the tab (only samples with a $c = 0$ — solvent-reference — curve appear: I·sin θ plots the $c = 0$ intensities).

> **Tip.** A flat I·sin θ is much harder to achieve with water than toluene. Overlaying normalized curves for both is an effective alignment/stray-light check.

9.3 Synthetic Data Generator

Generates artificial light-scattering data with **ground-truth** parameters — to validate the analysis (recover a known answer) or produce test files. The forward model uses the **same** q, Stokes–Einstein, optical-constant, Rayleigh, and Zimm physics as the analysis code, so loading and analyzing the

output recovers what you put in (no convention mismatch).

Inputs. *DLS* and *SLS* need different physical parameters, so they have separate input blocks that share the optical **conditions** (wavelength, n, temperature, viscosity) and the angle/concentration grid:

- **DLS sizes** — a populations table (median R_h, relative scattering weight, lognormal spread; one row = monomodal, two = bimodal, …) plus the coherence factor β, noise level, and point count. The populations *are* the modality — there is no separate "bimodal" option, and the CONTIN/NNLS distribution is something you then recover by analyzing the generated correlogram, not a generated artifact.
- **Lag axis and noise model** (in the same block) — how the correlogram is sampled and how noisy each channel is. *Log (uniform)* spaces the delay times evenly and uses the **Points** count. *Multi-τ* instead builds the staircase a real correlator produces: the sample time doubles every L channels, set by Δt_0, channels per octave, and the number of blocks. Every instrument this program reads emits a multi-τ axis, so it is the more realistic — and harder — test of a distribution fit. The choice applies to the **Correlogram** and **Multi-angle DLS** artifacts, which then ignore **Points**; every other artifact keeps a uniform grid and still uses it. The noise model does the same for the noise: *Constant* puts one standard deviation on every channel, while *Per-channel (Schätzel)* varies it the way a correlator does — quietest through the decay, noisiest in the coarsest blocks — with **Noise level** setting the first channel's SD. See the **Theory-and-Equations-Guide** §7 for both.
- **SLS sample** — M_w, R_g, A_2, and $\mathrm{d}n/\mathrm{d}c$. Untick *Write dn/dc into the SLS file* to produce a file that withholds the increment (the intensities are still generated with it): the program then analyses it on the arbitrary optical scale, as it would a real file with no $\mathrm{d}n/\mathrm{d}c$.
- **Depolarization / shape** (for the *Depolarized (VV/VH)* artifact) — a depolarization ratio ρ_v and a particle **shape** (rod or ellipsoid) with its dimensions, which set the depolarized VV/VH channels over the shared angle grid.
- **Calibration** — *Default* writes calibration metadata into a saved SLS file (and, when **Add to workspace as sample** is ticked, prefills the session calibration so M_w is absolute); *Uncalibrated* omits it, so an analysis run is flagged arbitrary-scale.

Artifacts. Tick **Preview** and/or **Save** per artifact:

Artifact	What it tests	Saved as
Correlogram (one angle)	cumulant / single / double / KWW / CONTIN / NNLS	generic two-column CSV
Count-rate trace	trace diagnostics (outliers, windowed Hampel, drift, Fano, ADF)	two-column CSV
Multi-angle DLS (all angles)	Γ-vs-q^2	multi-angle `.ASC` (+ a trace per angle)
SLS — full Zimm set	Zimm / Berry (thermodynamic)	multi-concentration SLS `.csv`
SLS — form factor P(q), sphere / coil	Form factor P(q) (large qR_g curvature)	multi-concentration SLS `.csv`
SLS — single concentration (angular)	Debye / Guinier (apparent)	SLS `.csv`

Artifact	What it tests	Saved as
SLS — single angle (conc. series)	calibration-free A_2 / single-angle M_w	SLS `.csv`
Depolarized (VV/VH)	static Cabannes (δ^2, f) + dynamic DDLS (D_r, shape)	two ALV `.ASC` (VV + VH)
Architecture series (linear + branched)	Cross-Sample branching g / h / arm-number n	a linear + branched folder
Form-factor series (SLS + DLS)	Cross-Sample ρ shape verdict, degenerate-solution selection, model-dependent $\nu \approx 1/3$ scaling, apparent / uncalibrated	an SLS + paired-DLS folder
Full M_w-series test set	everything (a homologous series)	a Clean + a Messy folder

Preview draws the *raw generated data* (the analysis figures come from loading the data into the analysis tabs). **Save** writes loadable instrument files into the chosen folder. **Add to workspace as sample** injects a coherent sample (the primary DLS artifact, the primary SLS artifact, and any trace) under a parsed `name / solvent` identity, so DLS + SLS connect for $\rho = R_g/R_h$ without touching disk. Preview never changes any state.

The **Depolarized** artifact bundles both depolarized-scattering inputs: a static VV/VH intensity pair (its ρ_v, δ^2, and Cabannes factor are shown — copy the two intensities into the SLS *Anisotropy (Cabannes) check*), and a dynamic VV + VH multi-angle set that, injected, is already tagged VV/VH so the DLS tab's DDLS sub-tab pairs it and recovers D_r and the shape. The **Architecture series** is a batch: injecting it adds a whole linear reference ladder plus branched samples, each already tagged **Linear** / **Branched**, so the Cross-Sample *Architecture* tab reads g / h / n out of the box.

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10. Running an Analysis

This section summarizes, procedurally, how to use the three analysis tabs. For the equations, derivations, and literature behind each method, see the **Theory-and-Equations-Guide**.

10.1 DLS tab

The DLS module extracts particle dynamics — decay rates, diffusion coefficients, and hydrodynamic radii — from a measured $g_2(\tau)-1$.

Procedure. Tick one or more measurements in the checklist, set the delay window and baseline, choose a method (or tick the distribution methods you want), click **Run**, and read the result in the table and plot. See the **Theory-and-Equations-Guide** for the equations and theory.

The tab is organized into **sub-tabs**, each a persistent full-size view (running one never clears another):

- **Correlogram** — fits the raw correlogram with a parametric model (cumulant / single / double / KWW), flagging anything the fit cannot honestly support.

  - **Scales.** The correlogram is shown in **four scales at once** — a large lin-log view plus lin-lin / log-lin / log-log thumbnails; **double-click a thumbnail to promote it** to the main view. A residual panel sits below, and **Cumulant** and **KWW** fits each populate their own results table. The raw correlogram appears as soon as you tick a measurement, before any fit.
  - **Cumulant order.** Set here (seeded from Settings; the per-run value wins).
  - **Cumulant method.** A global choice in **Settings ▶ Cumulant method**: *Nonlinear* (Frisken, the default — fits g_2-1 directly with a floating baseline, more robust to drift and noisy data) or *Linear* (Koppel — the classic log-polynomial fit). It applies to every cumulant-based step (cumulant, Γ vs q^2, replicate averaging); the table shows which method produced each result (and its fitted baseline). Switching the method clears existing cumulant-based results — you are asked to confirm first (the dialog says how many results, and how many picked values, will go; Cancel changes nothing) — and distributions, SLS, and any hand-entered R_h are kept. That set includes a **D vs c** extrapolation whose decay rates came from the cumulant fit; one built from the single-exponential fit does not depend on the method and is left alone. A sample R_h taken from a cleared result goes with it — including one you **picked** yourself, since the fit it named no longer exists under the new method — so $\rho = R_\mathrm{g}/R_\mathrm{h}$ is blank until you re-run or re-pick.
  - **Delay window.** Min/max, in µs, restricts which lags are fitted.
  - **Skip initial lag channels.** A separate global setting, **Settings ▶ Skip initial lag channels**, drops the first N channels (the shortest lags) from *every* DLS fit — set it to your correlator's afterpulsing count when a fit is dragged to a falsely small size by a noisy first channel (default 0; the skip and the delay-window minimum compose, the later of the two winning).
  - **No viscosity.** A measurement loaded **without a viscosity** still fits — decay rate and diffusion coefficient need no viscosity — but its R_h cells read *n/a* with a ⚠ whose flag line says why (Stokes–Einstein needs the viscosity); the exported R_h columns carry the same reason in their Comments cell.
  - **Undecayed modes.** A ⚠ also marks a decay that never finished inside the delay window: the fit reports a rate, but the measurement did not observe the relaxation, so that rate — and the size from it — is an extrapolation. For the exponential fits the mark is **per mode**, so a healthy fast mode beside an undecayed slow one keeps a clean number, and the flag line names which (*fast / slow / both modes*); the cumulant and KWW fits judge the correlogram as a whole. Measure a longer delay range; a static or aggregated sample is the usual cause.
  - **Other flags.** Every method also flags a size outside the range DLS can establish (above ~5 µm, or below 0.2 nm — a short-lag artifact; skip the leading channels), and the cumulant flags its R_h when the PDI exceeds 0.3 or the fitted decay rate is not positive. The same ⚠ and reason appear in the Summary tables and in the exported R_h Comments cells, and a flagged value is offered in the Cross-Sample R_h picker but never chosen for you.
- **Distribution** — recovers a full distribution of decay rates without assuming a functional form (tick one or more **methods** to compute, with the R_h/Γ axis toggle). NNLS resolves well-separated peaks but is spiky, CONTIN is smoother and generally better for broad/real data, and Lognormal is a robust single-mode default for a known-unimodal sample. A Lognormal fit that **fails to converge says so**: its legend entry reads *(failed)*, a red flag explains that the curve shows the starting estimate rather than a fit, and no peaks, Summary rows, or ρ

sources are offered from it (its export marks the affected columns' Comments the same way). Resolved **peaks are labeled** on the plot, and a lower panel — chosen with the **Bottom panel** selector — shows either the fit **residuals** (data − reconstructed g_2-1 vs delay time) or, for CONTIN, the **α-sweep** diagnostic behind the automatic regularization: the L-curve corner, the GCV minimum, or the F-test curve, with the chosen α ringed, and a red halo when it landed on a sweep edge. At the ceiling that means possible over-smoothing. At the floor it means the selector wanted a smaller α than the range allows; at a low floor such as the default 1e-4 the fit is then essentially the unregularized NNLS one, whose extra peaks may be noise. One sweep per CONTIN curve, colored to match the distribution; **Export α-sweep...** writes it to a CSV. The R_h grid (min / max / points), the **CONTIN α-selection** method (**GCV** by default — self-calibrating — or the L-curve corner or F-test), and the **CONTIN** α range are set here (seeded from Settings; the per-run value wins). The sub-tab has its own **Peak results** table (rows *1st peak / 2nd peak / …*, one color-matched column per ticked measurement·method, cells reading e.g. *"12.3 nm · 87%"* — the peak R_h and its intensity-weighted area), mirroring the Correlogram results panel; the same values also roll up into the workspace-wide **Summary** sub-tab. A **Weighting** selector re-expresses this same intensity-weighted result as **volume**, **surface**, or **number** — a re-presentation, not a re-fit, so switching it never recomputes the distribution and carries no ±. The **number** view divides out how strongly each size scatters, which magnifies small-size noise; ISO 22412 §9.1 deprecates number distributions from DLS for that reason, so treat a large small-particle population there as suspect. Reweighting is well-posed only for **spheres**; the **Sphere model** (Rayleigh / RDG / Mie) is chosen to match the sample, and RDG and Mie both need the solute refractive index entered on the Data tab. See the **Theory-and-Equations-Guide** §3.6.

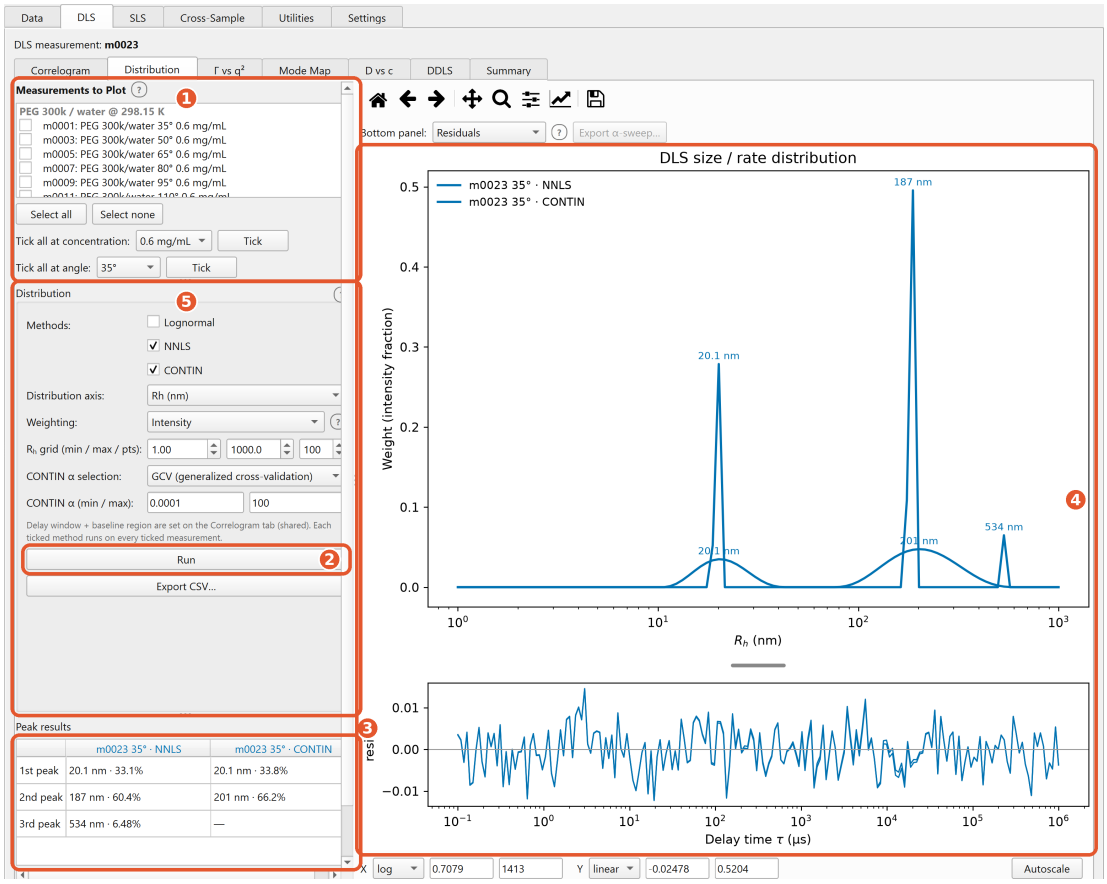


Figure 3: The DLS **Distribution** sub-tab on a bimodal sample. Tick the measurement(s) (1) and the methods in the **Distribution** box (5), then press **Run** (2); each method's size distribution is overlaid on the plot (4) with its peaks labeled, the residuals below (or, via the **Bottom panel** selector, the CONTIN α-sweep), and the resolved R_h and intensity fraction of every peak in the **Peak results** table (3). NNLS gives sharp peaks, CONTIN a smoother curve.

- Γ **vs** $\mathbf{q^2}$ and **D vs c** — sample-level analyses that test whether motion is genuinely diffusive and extrapolate the diffusion coefficient across angle or concentration.
  - **What each computes.** "Γ vs q^2 (sample, multi-angle)" tests whether motion is genuinely diffusive and returns D (and R_h); "D vs c (sample, multi-conc.)" extrapolates the apparent diffusion coefficient to zero concentration, returning the infinite-dilution $R_{h,0}$ and the interaction parameter k_D. Both operate on the selected sample's DLS measurements (of the current M_w fraction).
  - **Points table.** Each tab carries a **"Measurements (tick to include in the fit)"** points table — one row per measurement (Angle · c · Γ · D_{app} for Γ vs q^2; c · Angle · D_{app} for D vs c) — and the tick boxes choose which points enter the fit. On focus the table lists the sample's measurements as metadata only (Γ and D read "—"), and it **refreshes as soon as you commit** in the Data tab — no sidebar re-click needed. Every eligible row is ticked by default; changing the selection marks a shown result *"selection changed — press Run to refresh"* rather than refitting live or clearing the plot.
  - **Bulk helpers. Select all**, **Select none**, and **Tick all at** a chosen value — a dropdown of the distinct **concentrations (in mg/mL)** on Γ vs q^2 and **angles (in °)** on D vs

c. (The **Correlogram** and **Distribution** sub-tabs have both bulk selectors too.)

- **Ineligible rows.** Rows that can't be ticked are grayed with a **hover tooltip explaining why** (parameters not confirmed, or a different M_w fraction); a note also flags any same-polymer/solvent measurements held in a separate sample because their temperature isn't confirmed yet.
- **Run and results. Run** stays disabled until at least two distinct angles / concentrations are ticked (a status line reads *"Tick at least two distinct angles to run."*). After a Run the Γ/D_{app} columns fill in, a point whose cumulant fit failed or whose PDI exceeds the validity limit is flagged in its tooltip, and unticked points gray out on the plot. Scalar outputs fill a **Results** table (D, R_h, R^2, *Diffusive?* for Γ vs q^2; D_0, $R_{h,0}$, k_D, R^2 for D vs c).
- **How Γ and D are fitted here.** They come from an **internal 2nd-order cumulant fit** of each correlogram — independent of any saved Correlogram/Distribution result — using the global skip-channels and cumulant method from Settings.
- **Mixed polarizations.** Both fits **refuse a ticked set that mixes polarizations**, naming the tags found. A sample may hold both VV and VH runs — that pairing is what DDLS is for — but $\Gamma_{VH} = \Gamma_{VV} + 6D_r$ is a rotational term, not scatter, so one line through both fits neither. Tick a single geometry, or run DDLS. Runs that share one tag, or that are all untagged, are unaffected.

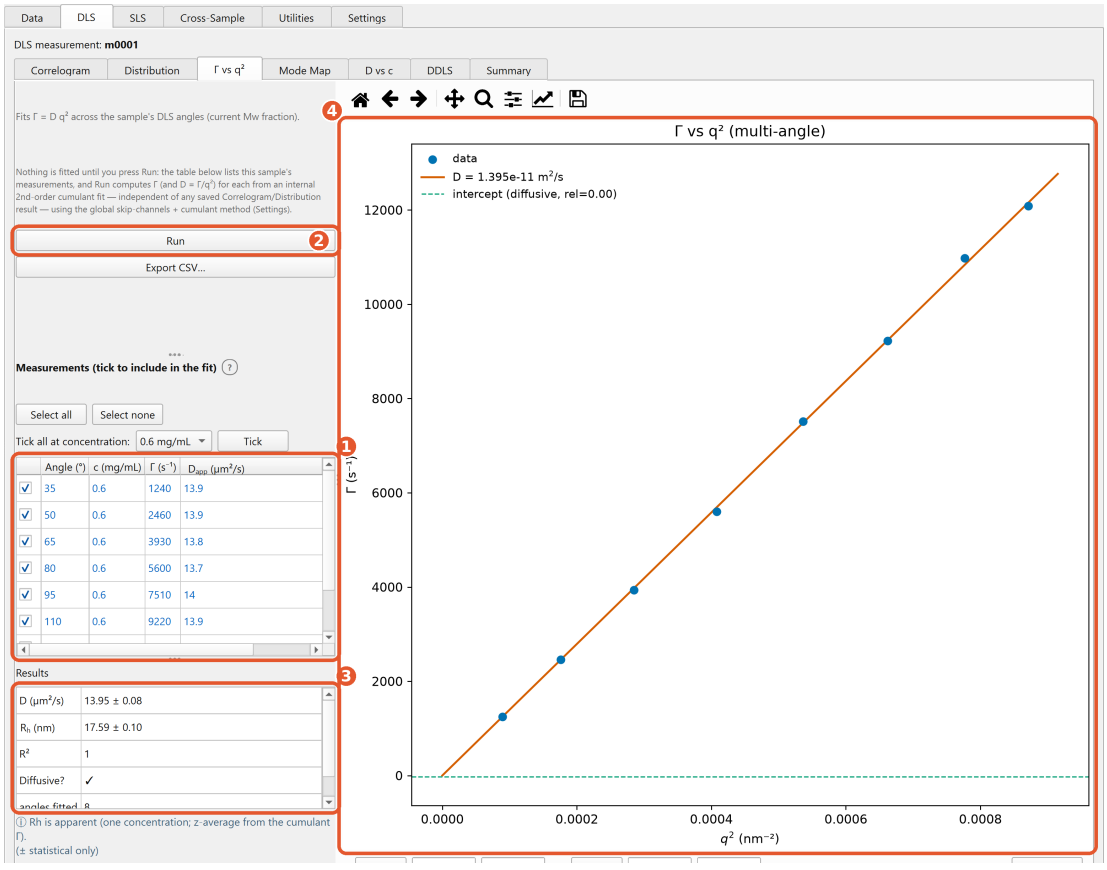


Figure 4: The DLS Γ **vs** $\mathbf{q^2}$ sub-tab. Tick which angles enter the fit (1) and press **Run** (2); the decay rate Γ is fitted against q^2 across the sample's angles (4). A straight line through the origin confirms diffusive motion; the results table (3) reports D, R_h, R^2, and the diffusive check.

- **Mode Map** — the mode-resolved companion to Γ vs q^2, for samples that relax on more than one time scale (a fast + slow mode).
  - **Run and Source. Run** extracts *every* mode of *every* ticked run — with the chosen **Source** (the two-mode double-exponential, or a CONTIN/NNLS distribution's resolved peaks) — and plots them all in (q^2, Γ) space. Both distribution sources invert on the **global R_h grid from Settings**, and CONTIN also on the α **selection** there (NNLS is the unregularized case and has no α) — the same settings the Distribution tab seeds its per-run fields from; a denser grid resolves more closely-spaced modes and costs sharply more time per run. The Distribution tab additionally applies its own delay window and baseline, so a run fitted there can resolve differently from the same run on this map.
  - **Families.** The program deliberately does **not** pair modes across angles: **you** do, by adding a **family** (Families panel), selecting it, and **clicking map points into it** — click again to remove, click a point held by another family to move it (the status line says which happened). Each family with points at ≥ 2 angles is fitted on the spot: through-origin D with an **apparent, mode-resolved R_h** (always computed, always labeled — *whether the family passes through the origin is a label with stated tolerances, never a filter*), plus the slow-mode deviation form $\tau_s^{-1} = aq^2 + bq^4$.
  - **Marker warnings.** Two independent warnings ride on the markers, one per channel so they compose: a **hollow** marker flags a **low-amplitude mode** (under ~2 % of the run — imprecise), and a **square** marker flags a mode whose **decay never finished** inside the delay window (its rate is extrapolated). A hollow square is both.
  - **Polarization.** The map shows **one polarization at a time** (VV / VH / unpolarized); a family can never mix them.
  - **Role and Amplitude view.** Designate one family **fast** and one **slow** (the Role drop-down) and switch the View to **Amplitude vs q** for the A_f/A_s vs $\sin^2(\theta/2)$ report with its angle-flatness note.
  - **Persistence.** Picks are saved with the session and matched to modes **by value** — re-running the extraction (or switching Source) never re-aims a pick; orphaned picks are counted in the Families panel and named on the fit.
  - **Uncertainty.** Mode amplitudes and per-run Γ carry **no ±** (single-correlogram quantities); the per-family D, R_h, a, b carry statistical SEs.
  - **Export. Export CSV...** writes the per-point map (with each point's family), every family's fit with its labels in the Comments cells, and the amplitude report.
  - **Try it without data.** Utilities ▶ Synthetic Generator ▶ **Two-mode DLS (fast + slow, crossing)**.
- **DDLS** — pairs a VV with a VH correlogram at each angle (tag each correlogram's polarization in the Data tab) to separate translational from rotational diffusion, and fits sphere and rigid-rod shape models for particle dimensions.
  - **Paired Angles.** A checklist ticks which paired angles enter the D_r/D_t fit (all ticked by default; untick an outlier), and *Include all angles* re-ticks them. Press **Run** to fit; the sphere model needs the solvent viscosity.
  - **Repeats.** Several VV (or VH) correlograms at one angle are treated as repeats and averaged, so DDLS first checks that every correlogram it is about to combine is from one sample and condition, and **refuses the run** if the concentration or M_w fraction differs. Worth knowing because a concentration series is deliberately one sample in the sidebar, so a second concentration arrives here looking like a repeat.
  - **Results.** The rotational-diffusion results and the sphere/rod shape-model outputs each fill their own **results table**.

  - **Unphysical D_r.** If any angle comes back with $\Gamma_{VH} \leq \Gamma_{VV}$ the D_r is unphysical and a warning says so; **Export CSV...** repeats that warning in the Comments cell of every column whose value depends on D_r — including the rod, sphere and ellipsoid dimensions, which one bad angle can corrupt while leaving the average positive — so it reaches Origin with the number rather than staying on screen.

  Across all of the DLS analysis tabs above: if you go back to the Data tab and **commit a change to a parameter that fed a fit already on screen**, the plot stays but a status line reads *"Inputs changed since this fit — press Run to refresh"* — so a result is never silently shown against parameters it wasn't computed with. The same hint appears when any other input of the shown fit moves: the delay window or baseline (dragged or typed), the cumulant order, the ticked set, or a Settings change to the cumulant method, the uncertainty estimator or *Skip initial channels*. The hint is specific to the results whose own inputs changed. (The **SLS tab behaves the same way** — switching sample/fraction or committing redisplays the last fit and flags it stale; only hiding/showing a masked point refits on its own. See §10.2.) A fit that finishes after you have clicked another sample or another M_w fraction is not painted over the new view; it waits in its own slot and shows when you focus that sample and fraction again.

- **Summary** — a workspace-wide, exportable results table that **persists across save/reload** (§8.2).
  - **Per-measurement table.** One row per measurement shows cumulant R_h
    - PDI, single/KWW R_h, and NNLS/CONTIN peaks as `Rh (Int %)` (Int % is the intensity-weighted peak area, *not* a mass/weight percent).
  - **Sample-level table.** Collects replicate averages, Γ–q^2 (q→0, **apparent**) and D–c (c→0, **thermodynamic**) — the R_h Type column marks that distinction and **From** lists the contributing measurements. A cumulant R_h outside the size range DLS can establish is marked ⚠ — hover the cell for the reason, and see §3.1 of the **Theory-and-Equations-Guide**. The exported CSV carries the same qualification in the R_h column's Comments cell, naming the affected rows.
  - **Filtering and export.** The left checklist is shared with the Correlogram/Distribution tabs and doubles as a **"Ticked only"** filter; **Export CSV...** writes one long/tidy row per result or peak.
  - **How rows fill and empty.** Rows fill in as you run fits and averages (e.g. *Average derived results* from the sidebar). A row can also go away: if you average the same method again and the fits come back with an answer that cannot be used — sizes outside the range DLS can establish, or a mean that overflowed — the earlier replicate average is removed rather than left standing, and the sample R_h goes with it when it came from that average. The summary box names the value it dropped. If the fits produce no usable result at all — too few of them converged, or none did — nothing is removed, because a run that settles nothing says nothing about the average before it.
  - **Your own picks.** A value you entered or picked yourself is never removed — and when you have picked one, re-averaging still refreshes the row, so the table shows the average you last ran rather than the one that predated your pick.

Co-plotting several measurements. Both the Correlogram and Distribution tabs carry a **"Measurements to plot"** checklist of every loaded DLS measurement (any sample, angle, concentration, or M_w fraction); tick several to overlay them for comparison. This checklist — **shared between the two tabs** and mirrored in the Workspace list (ticked measurements read bold blue) — is the **sole driver** of what gets fit; selecting a measurement in the Workspace list only *navigates*.

Run fits every ticked measurement (and, on the Distribution tab, every ticked **method**). Fitting many heavy distributions (especially CONTIN) at once is slow, so tick only what you want to compare.

Every single-axes DLS/SLS analysis plot carries an **axis-control bar** beneath it: per-axis linear/log toggles, explicit min/max boxes, and an autoscale button.

> **Delay-time window + baseline region.** A single shared analysis region (set on the **Correlogram** sub-tab and used by all per-measurement DLS methods, on every ticked measurement) defines the fit window and the baseline region. Set them by typing values or by **dragging the markers on the correlogram**: the green dotted lines are the window (points outside are ignored — useful to trim a noisy long-delay tail), and the purple dashed/shaded span is the **baseline region** used by the distribution methods (parametric fits don't use a baseline). When a window and a baseline marker overlap, tell them apart by their **offset caret handles**: the **window** handles are **green carets along the top** of the plot, the **baseline** handles are **purple carets along the bottom** — grab the top or bottom half of the plot to pick the one you want. Defaults: full range (the union of all ticked measurements), and the last 25 % of delays for the baseline. A ticked measurement whose lag axis doesn't overlap the window (e.g. a millisecond-lag set co-plotted with a microsecond-lag one) can have too few points inside — it is then **skipped with a note**; drag the markers to a suitable range.

10.2 SLS tab

The SLS module extracts the weight-average molecular weight M_w, the radius of gyration R_g, and the second virial coefficient A_2. It first converts raw intensities to absolute excess Rayleigh ratios (via the calibration panel), then constructs the Zimm, Berry, or Debye plots. Pick which sample to analyze from the **SLS sample** dropdown at the top of the tab (only samples with SLS data appear); the whole Zimm set — solvent reference plus the concentration series — is analyzed together. This **Analysis** sub-tab is one of three: the **Derived Intensity** sub-tab (§10.2.1) builds a series from DLS runs when none was measured directly, and the **Mode-Resolved** sub-tab (§10.2.2) partitions a series by DLS mode amplitudes.

Procedure. Set the calibration in the panel (or accept the uncalibrated fallback), choose the fraction (if any), select a method, mask any bad points/angles, click **Run**, and read M_w / R_g / A_2 with their apparent/thermodynamic and calibrated/uncalibrated flags. See the **Theory-and-Equations-Guide** for the equations and theory.

Calibration panel. The program computes its own calibration constant k_c from a single calibrant measurement — the intensity of a standard liquid (usually toluene), the angle it was measured at, and that standard's geometry-aware Rayleigh ratio. The constant is shown and can be edited. Any value from a data file is kept only for reference and not used in the calculation. If no k_c can be supplied, the analysis still runs on an arbitrary scale, flagged uncalibrated — then M_w and the absolute A_2 are marked unreliable, but R_g and the calibration-free product $2A_2M_w$ remain valid.

The standard's refractive index is **proposed** from the solvent library for toluene at the standard's wavelength and temperature, and re-proposed when you change either, until you type a value of your own. A **teal dot** beside the field marks the library's proposal, as on the Data tab; a value you type has no dot and is never overwritten, and clearing the field returns it to the library value. The field is **blank** only where the library has no value for the standard's condition, and a blank

is not the same as leaving the correction out: the $(n_{\text{solvent}}/n_{\text{standard}})^2$ correction is then unapplied, and the run is flagged uncalibrated exactly as a missing k_c is. A note beside k_c says, before you run, when the standard temperature or wavelength is off the toluene Rayleigh-ratio table (the value is extrapolated), when the standard wavelength is not the wavelength your SLS data were measured at, and when the standard refractive index is blank or differs from the library.

Methods available.

- **Single-concentration (apparent) Debye** — at one concentration, gives an apparent M_w and apparent R_g. A single angle yields only an apparent M_w and no R_g. Apparent values are never presented as final.
- **Zimm / Berry double extrapolation** — combines several concentrations and angles and extrapolates to both zero angle and zero concentration, giving M_w, R_g, and A_2 at once. The result is marked **thermodynamic**. Berry linearizes the high-qR_g regime (large M_w) better than Zimm; the program reports which construction it used.
- **Form factor P(q)** — fits the **full angular form factor** instead of the small-qR_g slope, so it stays valid to larger qR_g where the Zimm/Guinier plots curve. A **P(q) model** chooser picks a **Sphere** (which also reports the geometric radius R) or a **Gaussian coil**. One concentration gives an apparent result; several give a thermodynamic M_w, R_g, and A_2. As with the other routes, R_g (and R) survive an uncalibrated run.
- **Calibration-free A_2** — obtains the product $2A_2M_w$ from ratios of excess intensities across concentration at a fixed angle, without any absolute calibration. Valuable when absolute calibration is uncertain.
- **Guinier** — a single-concentration route to R_g from the log intensity, scale-independent and reliable even on an uncalibrated run.
- **Rayleigh series** — the excess Rayleigh ratios themselves.

  Spacing constant (Zimm/Berry grid). Set the purely **aesthetic** spacing constant in the **Spacing k** box (or press **Suggest** for the automatic value). It only changes how spread-out the curves look — M_w, R_g, and A_2 come from the real fit and are unaffected.

Data masking. You can hide whole angles/concentrations or individual points (click to mask, grayed and persisted); an over-masked Zimm is redirected to Debye/Guinier. A manual-M_w override is available — useful when the trustworthy M_w was characterized independently (say, in a different solvent or by GPC/MALS) and you want the analysis to use that value — and it is never overwritten by a re-analysis.

Anisotropy (Cabannes) check. For an optically anisotropic scatterer the depolarized scattering inflates the apparent M_w. Tick **Anisotropy (Cabannes) check** in the **Calibration** panel and enter the depolarization ratio ρ_v — directly, or as I_{VV} and I_{VH} intensities. It reads out ρ_v, the optical anisotropy δ^2, the Cabannes factor $f = 1 - 4\rho_v/3$, and the isotropic-corrected weight $f{\cdot}M_w$ for the sample's analyzed M_w. The first three are ratios and always show; $f{\cdot}M_w$ is a mass, so an uncalibrated run dashes it out like every other absolute M_w — unless the M_w is one you entered by hand, which is trusted and still corrected. This is **display-only** — nothing is written to the sample or exported, so a corrected M_w can never silently propagate into Cross-Sample or a saved file. If you trust the corrected value and want it used downstream, enter it by hand as a **Manual M_w** (Analysis). (For a flexible coil $\rho_v \approx 0$ and the correction is negligible; it matters for rods and strongly anisotropic particles, where R_g is also anisotropy-distorted — see the **Theory-and-Equations-Guide** §4.6.)

Results. Each method fills a two-column **Results** table — Zimm/Berry give M_w, R_g, A_2 and R^2;

Debye/Guinier the apparent values; single-angle an apparent M_w; calibration-free A_2 the $2A_2M_w$ product. The Rayleigh series has no scalar table — it shows a one-line status (*"Excess Rayleigh ratio for N concentrations."*) alongside its plot. **Run is the only thing that computes a fit** (the same explicit-Run model as the DLS tabs): switching sample, fraction, or method **keeps the last result on screen** rather than blanking it, and if you commit a parameter that fed the shown fit a status line flags it *"Inputs changed since this fit — press Run to refresh"* instead of silently recomputing. The one exception is **masking** — hiding or showing a point re-runs the fit immediately.

10.2.1 Derived Intensity sub-tab Build an SLS intensity series directly from this sample's DLS runs, so a goniometer that recorded only correlograms still yields a full static analysis. Each DLS run contributes its mean count rate as the intensity at its angle, and runs at the same angle and concentration are averaged into one measurement per concentration.

Procedure. Tick the DLS runs to use (**Select all** / **Select none** help with a long list), optionally enter a $\mathrm{d}n/\mathrm{d}c$ for the derived points, and click **Derive**; the created measurements appear on the **Analysis** sub-tab like any other SLS data.

- The table lists each run's **Angle**, **Count rate**, **Source**, concentration, and a **Status** of *ready* or *not usable* (a not-usable row's reason is its tooltip).
- The mean count rate and its source are set on the **Data** tab and shown here read-only — there is exactly one place that number is chosen.
- Derived points are ordinary SLS measurements: calibrate and analyze them on the **Analysis** sub-tab like any loaded series.
- Their intensities are count rates in cps. The calibrant intensity and any solvent blank analyzed with them must be in cps as well — only the ratio matters, but the units must match.
- A derived point is a **snapshot**: re-running a fit or editing a source run never changes it. Click **Derive** again to refresh it.

See the **Theory-and-Equations-Guide** §4.1.1 for how a derived intensity series is built and analyzed.

10.2.2 Mode-Resolved sub-tab Split an SLS intensity series by the amplitudes of this sample's recorded DLS modes (the Sedlák partition) and analyze each mode's share of the scattering on its own.

Procedure. Record a mode set on the DLS tab's **Mode Map** first (**Record mode set**); select it here, click **Partition**, then pick a **Mode** and an **Analysis** method and click **Run**.

- A mode set appears here as a read-only snapshot with its method, regularization, and recording date — re-fitting on the DLS side never changes it; record again for a fresh one. **Open in Mode Map** jumps to where it was defined.
- **Partition** splits every SLS series of this sample (loaded or derived) that shares a concentration with the set: each mode's share is $\mathrm{d}R_{\mathrm{mode}}$ = amplitude fraction $\times$ $\mathrm{d}R$.
- **Amplitude Guinier** fits one mode's series alone and gives that mode's apparent size.
- Guinier, Debye, Zimm, and Berry run on a mode's series like any SLS analysis; every result stays flagged apparent and mode-resolved, with the partition recorded in its export Comments. A per-mode series is never added to the workspace as its own measurement, and a mode's M_w is never written to the sample — it describes that mode's share alone.

See the **Theory-and-Equations-Guide** §4.1.2 for the partition and what each per-mode analysis assumes.

10.3 Cross-Sample tab

The Cross-Sample module computes quantities that link samples or modules.

Procedure. Include the samples you want, pick the focused sample/fraction with the Sample and Fraction selectors, choose the R_g / R_h / M_w / A_2 source per row, read ρ and its Shape label in the table, and switch to the Scaling sub-tab for the log–log plots and fitted exponents. See the **Theory-and-Equations-Guide** for the equations and theory.

It is *aggregate-scoped*, so it makes **two** selections, both in the tab:

- **Membership** — an **include/exclude** list (left) of every sample with SLS data (all start included); untick to drop a sample from the ρ table and scaling plots.
- **Focus** — the source panel's own **Sample** and **Fraction** selectors, which say exactly which sample and M_w fraction you are editing. This is explicit: the shell Workspace tree only *highlights* the focused sample (read-only) — clicking it there does not drive this tab, and the ρ table is a read-only results view. The Fraction list holds every fraction the table shows for that sample — a label carried by the SLS series and one carried only by DLS measurements alike — so every row can be pointed at.

$\rho = R_g/R_h$. The shape parameter links R_g (from SLS) and R_h (from DLS) of the same sample. Each included sample (or *fraction*) gets one row: R_g, R_h, ρ, a **Shape** label naming the likely architecture, and whether ρ is **apparent** or **thermodynamic**. The program checks that the two measurements share polymer, solvent, and temperature.

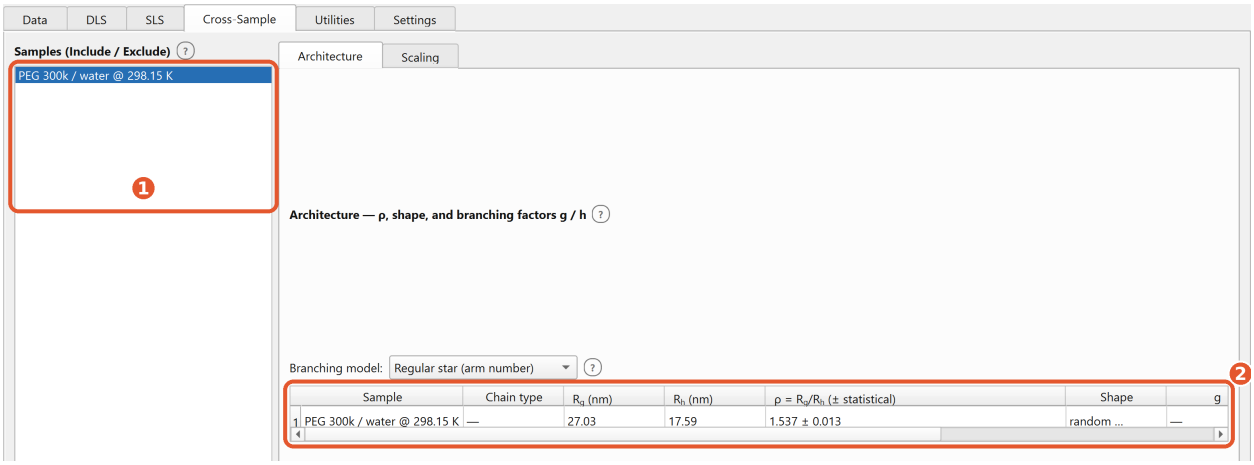


Figure 5: The Cross-Sample $\rho = R_g/R_h$ table. Tick which samples to include (1); each included sample gets one row (2) with its R_g, R_h, ρ, a Shape label, and an apparent/thermodynamic flag. A per-row dropdown (not shown) chooses the source for each quantity.

Because a sample can have several R_g and R_h values, you **choose the source** for each (a dropdown per row), and the program never picks silently:

Quantity	Sources (best-tier → apparent)	Hand-entry?
R_g (SLS)	Zimm/Berry double extrapolation (*thermodynamic*); single-concentration Debye/Guinier (*apparent*); a **form-factor P(q) fit** (sphere/coil/ellipsoid/Mie/core–shell), flagged *model-dependent* (see below)	Yes
R_h (DLS)	$c\to0$ D-vs-c extrapolation (*thermodynamic*); $q\to0$ Γ-vs-q^2 at one concentration (*apparent*); **replicate average** (*apparent* — the one dynamic R_h carrying a real ± SE, so preferred over a single cumulant and never overwritten by a refresh); single-condition cumulant (*apparent*); CONTIN/NNLS distribution peak (*apparent*)	Yes
M_w (SLS)	Zimm/Berry extrapolation; single-concentration Debye/Guinier; a **calibrated** form-factor P(q) fit (flagged *model-dependent*) — all **calibration-dependent** (an uncalibrated fit is on an arbitrary scale, flagged below)	Yes
A_2 (SLS)	Zimm/Berry double extrapolation only (no single-concentration A_2) — **calibration-dependent**	No (picker only)

A_2 has **no hand-entry** because it is a solvent/temperature-specific interaction coefficient with no external "standard", and it is the very quantity the A_2–M_w plot fits. A hand-entered R_g, R_h, or M_w is never overwritten by a later re-analysis. A source you explicitly pick here is firmer than the auto-selected default, but not permanent: a passive refresh never touches it, while the two runs that write sample values replace it — a **Zimm/Berry** run replaces a picked $R_g/M_w/A_2$, and **averaging replicates** replaces a picked R_h; the run reports the replacement, and **Settings ▶ Ask before a Run replaces a picked value** adds a confirm if you want one. Every other run offers its new fit in the picker instead of replacing your choice. The two Settings switches that invalidate fits wholesale (**Cumulant method** and the **Regression SE estimator**) clear a pick along with the results it came from, after you confirm.

Each picker groups its candidates by result type (extrapolated / distribution / averaged / single-condition) and hides the long tail of per-angle / per-concentration *apparent* values behind a **"Show all single-condition results"** checkbox. The best-tier value stays selected either way.

Model-dependent form-factor sizes. If you have run a full **form factor P(q)** fit (sphere, Gaussian coil, ellipsoid, Mie sphere, or core–shell) on the SLS tab, its R_g — and, when the run is calibrated, its M_w — also appears as a source, under its own *(model-dependent)* group and marked with a neutral information glyph. Unlike the slope-based R_g (Zimm/Guinier), a form-factor R_g assumes a particle **shape**, so it is right only if that shape is right. It enters at the **same tier** as its slope-based sibling (a multi-concentration fit is thermodynamic, a single-concentration one is apparent), but the program **never auto-selects a form-factor size over a slope-based one at the same tier** — you pick it deliberately. The flag travels into the ρ table (a neutral marker beside R_g) and the exported CSV (an **Rg source** column). Note that A_2 is **not** offered from a form factor: at $q\to0$ every shape converges to the same limit, so a form-factor A_2 would just duplicate the Zimm value.

Degenerate fits — choosing a solution. Ellipsoid and core–shell fits are often **degenerate**: more than one shape (prolate vs oblate, or different core/shell radius pairs) reproduces the same scattering curve, so the data alone cannot pick one. Each admissible solution is offered as its

own pick under a *(model-dependent — degenerate)* group, **labeled by its distinguishing shape parameters** — an ellipsoid solution by prolate/oblate, axial ratio, and R_g; a core–shell solution by its core and outer radii. Selecting one is **guided manual entry**: if you have independently resolved the shape (by electron microscopy, SAXS/SANS, or optical microscopy), you choose the matching solution; the program records *which* one you picked and flags every downstream value (ρ, the scaling exponent, the g/h branching factors, and the exports) as resting on a user-selected degenerate solution. The program **never** picks a degenerate solution for you — it is always an explicit choice. When a fit's solutions **agree on** R_g (within about 10%, or 1 nm) even though the shape is not pinned down — the size is determined, the shape is not — the picker additionally offers a single **"size determined, shape ambiguous"** value (the mean of the solutions' R_g), which *can* be selected automatically so you need not choose a shape. A contrast-matched core–shell with no defined R_g still offers none.

A form-factor source appears here **only while its fit is current** — computed against the same mask, calibration, and settings the sample carries now. If you change any of those after fitting (mask a point, edit the calibration, switch the uncertainty estimator), the form-factor option quietly disappears from the picker until you **re-run the fit on the SLS tab** (which also flags it stale there). This keeps a pooled form-factor size honest: it always reflects the current data, never a fit left over from earlier inputs. Re-opening a **saved session** (§8.2) behaves the same way: a form-factor R_g you had already **picked** is kept (its value and *model-dependent* flag travel with the sample), but the form-factor option itself is offered again only after you re-run the fit — the fit is recomputable state, not stored in the session file.

Calibration flag. M_w and A_2 are on an arbitrary scale until the run is calibrated, so each row shows an **"uncalibrated"** warning badge and the panel a one-line banner when the chosen value is uncalibrated. R_g is scale-independent and is never flagged. For an *SLS-only* sample the R_h (DLS) row is shown but disabled.

Picking a peak. If you have run a CONTIN or NNLS distribution for a measurement in the DLS tab, each resolved population appears as its own R_h source in the dropdown, so you choose which population feeds ρ. The dominant peak can be the default; a minor peak is selectable but never chosen automatically. The percentages are intensity fractions, not number or mass fractions. Distribution peaks are surfaced only from a distribution you have already computed.

The **default** source is the most-extrapolated the data support (a labeled choice), with fit quality as the tiebreak. ρ **is flagged apparent if *either* input is apparent.** For a physically meaningful shape parameter, prefer the thermodynamic (infinite-dilution) source on both axes.

Architecture sub-tab. Tag samples **Linear** (at least two, at different M_w) and **Branched** with the **Chain type** control on the Data tab: the linear R_g–M_w and R_h–M_w laws are fit to the Linear samples and read at each Branched sample's M_w. $g = (R_{g,br}/R_{g,lin})^2$ needs R_g; $h = R_{h,br}/R_{h,lin}$ needs R_h; both below 1 mean a more compact chain, and a star gives $h > g$. The **Branching model** selector sets how g is read — a regular-star **arm number**, or a random tri-/tetra-functional **branching density n** (the **n (branch)** column) — and is model-dependent: the same g reads as a different n under a different model. A cell reads — where it cannot be computed (hover for why); select a row for its readout. A ± appears only from replicate branched measurements. See the **Theory-and-Equations-Guide** §2.4–2.5.

Units. The R_g/R_h columns and the manual-entry fields follow the global **Settings ▶ Display Units** choice (e.g. nm or µm for size, g/mol or kg/mol for M_w); the scaling-plot axes use the same

setting, and switching either re-displays the same values without changing any fitted or reported number. A value you type is read in the shown unit.

Scaling laws (R_g–M_w, A_2–M_w). The **Scaling** sub-tab shows two stacked plots over the included samples (one point per sample/fraction), using each sample's effective M_w (a hand-entered M_w wins) chosen in the same source panel. The lower plot is always A_2–M_w; the upper plot is a **size** scaling whose axis you swap between R_g–M_w (SLS) and R_h–M_w (DLS) with the **Show R_h–Mw / Show R_g–Mw** button. Below each plot the tab prints a plain-language reading of the fitted exponent. Two cautions the tab surfaces: a plotted value can be on an **arbitrary scale** — an uncalibrated run, a missing dn/dc, or both — in which case the caveat names which quantity it could be. On a size plot that is the M_w, so the horizontal position is the meaningless one and a trusted M_w is the fix; on the A_2 plot it may instead be the A_2 itself, where the vertical position is affected and calibrating (or supplying a dn/dc) is. Either way the fitted **exponent survives** — a scale factor shifts the log–log intercept, not the slope — so it is the plotted values and the prefactor that are not absolute. The caveat appears whenever such a point is drawn, including on a plot too sparse to fit a slope. The second caution: A_2 can be **negative** (below the θ-point), in which case that point cannot sit on the log A_2 axis and is excluded. Curating which samples form the series is the user's responsibility — mixing polymers or solvents on one fit is not meaningful.

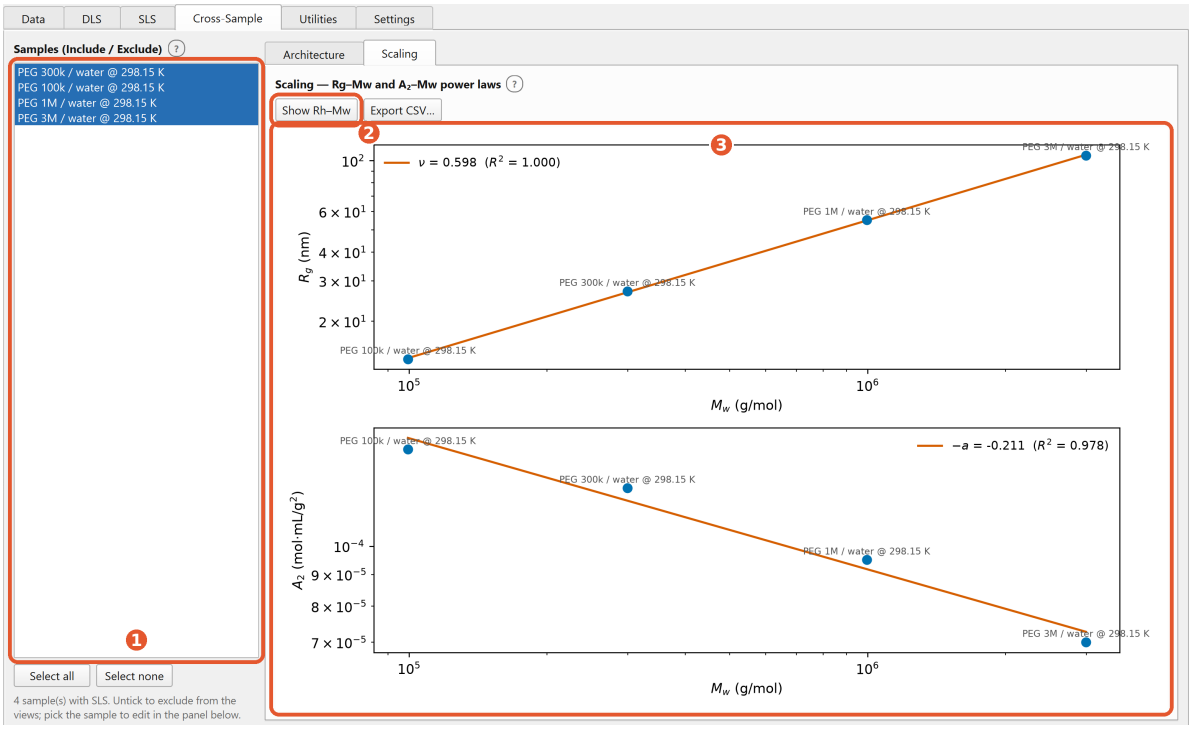


Figure 6: The Cross-Sample **Scaling** sub-tab over a PEG molecular-weight series. Each included sample (1) is one point; the upper plot fits R_g–M_w (swap to R_h–M_w with the button (2)) and the lower plot A_2–M_w, both on log–log axes (3). The fitted exponent is printed on each plot — here ν = 0.593, the value expected for a flexible coil in a good solvent.

11. Exporting Results

Every analysis result can be written to a CSV that OriginLab imports directly. Each file carries the four Origin label rows — **Long Name, Units, Comments, Parameters** — so axis labels and units populate on import. Tabular outputs (correlogram fits, distributions, Rayleigh ratios) are written column by column with units; scalar results (M_w, R_g, Γ, …) as single-value labeled columns. Zimm datasets use a wide layout (one $Kc/\Delta R_\theta$ column per concentration, the concentration in the Parameters row); its q^2 column is the union of the concentrations' angles, so where they do not share one grid each ordinate still sits on the row of the q^2 it was measured at, and a blank cell there means no value was written — that concentration has no point at that angle, or its ordinate is not a usable number (a masked or dropped point). Missing values are empty cells; a flag column that was evaluated (*PDI valid*, *Is diffusive*, *Guinier valid*, *Is apparent*, …) reads `yes` or `no` — blank there means the flag has no value at all (for example *Single-exponential valid* when no rod length was given). An uncalibrated result writes "uncalibrated, arbitrary scale" into the **Comments** cell of the affected columns only — their SE columns included — never changing the row count, so the Origin import stays intact. Every SE column's Comments cell says what kind of ± it holds ("statistical (excl. calibration/dn-dc)") and, when Settings selects the classical-OLS estimator, names it ("SE: classical OLS").

From the interface. Each analysis view has an **Export CSV...** button that writes the result currently on screen: the DLS **Correlogram**, **Distribution**, Γ **vs** q^2, **D vs c**, and **DDLS** subtabs (the DDLS file carries the per-angle Γ_{VV}/Γ_{VH} and D_r, the combined $D_t/D_r/R_{h,t}/\tau_{rot}$, and — when available — the rod, sphere and ellipsoid shape dimensions, each flagged "MODEL … not measured" in its Comments cell, the rod and ellipsoid ± cells also stating their Monte-Carlo census — how many of the draws a shape reproduced, and the seed); the SLS **Analysis** panel (the button exports the last method you ran — Zimm/Berry, Debye, Guinier, single-angle, calibration-free A_2, or the excess Rayleigh series); and the Cross-Sample **Scaling** tab (the R_g/R_h–M_w and A_2–M_w points and fits — branched points in their own columns beside the fitted ones, and (when any were dropped) an *Excluded* column counting the samples dropped before the fit). The button is enabled once a result has been computed. When a view holds several curves at once (an NNLS+CONTIN distribution overlay, or both size and A_2 scaling plots), the export writes one file per curve, suffixed by method/quantity; if any of those files already exist, the program lists them and asks before overwriting.

The save dialog proposes a filename that names the sample and the analysis — `PVP-water-298.1K_debye.csv` — built from the committed polymer, solvent, and temperature (sample-independent exports keep a plain name, e.g. `dls_summary.csv`). Characters that are not allowed in filenames are replaced with `-`. The dialog opens in the folder of your last export (your Documents folder on first use), and the plot toolbars' save button shares that folder and naming, so a plot image lands next to its CSV with a matching name. The open-file dialogs likewise remember the folder you last loaded data from, separately from the export folder.

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12. Visualization

Every plot shows exactly what was computed, nothing re-derived for display: correlogram fits with residuals, size distributions, Γ-vs-q^2 diagnostics, Debye plots, and the full Zimm/Berry grid. Distributions from different methods can be overlaid (e.g. NNLS and CONTIN on one axis), making their trade-off visible at a glance.

Data-quality and reliability **flags are interface overlays**, not part of the figure: a saved plot image is always clean. Where a result is apparent or uncalibrated, the interface marks it next to the plot, so a number can't be read off a chart without its caveat.

Display units (Settings tab). Plot axes follow **Settings ▶ Display Units** across the analysis tabs. The **Scattering $\mathbf{q}^2$** choice (nm^{-2} or m^{-2}) applies to every q^2 axis — the DLS Γ-vs-q^2 and DDLS plots and the SLS plots (Rayleigh series, Debye, Guinier, form factors, and the Zimm grid, whose printed spacing constant k follows the same unit). The **Concentration** choice governs the D-vs-c and calibration-free A_2 axes; legend concentration labels always read mg/mL. Switching either setting redraws the axes and labels only — no fitted or reported number changes with the unit choice.

Appearance (Settings tab). The **Theme** is *System* (follows your OS light/dark setting) by default; choosing *Light* or *Dark* explicitly **overrides** the OS so the application stays in that mode regardless of system preferences. The **Plot palette** (Okabe–Ito colorblind-safe / tab10 / grayscale) applies to plots drawn afterwards. Within the analysis tabs, the **stacked plots and their control columns are resizable** — drag the **splitter grips** (the visible three-dot handles) between them, including the fit/residual divider, which carries a persistent grip line. The workspace panel on the left is **resizable** (drag the divider) and long sample names show in full on hover; the window itself can be freely resized (tab contents scroll when the window is small).

See the **Theory-and-Equations-Guide** for the equations and derivations behind every plotted quantity.